

Nathaniel Rogers

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EDUCATION

Northwestern University, Evanston, IL

Anticipated Graduation: *March 2027*

Bachelor of Science in Biomedical Engineering | Segal Design Certificate

GPA: 3.94/4.00

Relevant Courses: Interdisciplinary Product Design, Design Thinking and Communication, Design Research, Biomedical Signals & Circuits, Systems Analysis, Robotics, Quantitative Systems Physiology, Bioelectronics, Biomedical Statistics

SKILLS

Hardware: 3D Printing, Tolerance Analysis, Rapid Prototyping, Circuit Design/Assembly/Debugging, Soldering, FMEA

Software: SolidWorks, OnShape, Python, MATLAB, Office 365, LTSpice, Arduino IDE, HTML, CSS, G*Power, SwiftUI

Communication: Project Management, Public Speaking, Presentation, Technical Writing, Motivation, Time Management

WORK EXPERIENCE

Biomedical Engineering Intern, Garrett Technologies, Inc.

June 2025 – August 2025

- Executed reliability testing on AED charging loops in an environmental chamber (50–60°C), diagnosing PWM instability under thermal stress and informing a firmware update improving performance at rated limit.
- Developed a Python-based tool to decode and visualize ECG, treatment, and AED system logs, generating automated PDF summaries with plots and error translations to reduce analysis time by ~15 minutes per log for a team of 9 engineers.
- Collaborated with internal engineering teams and external battery vendors to define acceptance criteria for lithium cell performance under dynamic loads and elevated temperatures, improving supplier reliability.

Ambassador, Vera Health

November 2025 – December 2025

- Onboarded 10 new users to a medical literature AI search engine within 3 hours at radiological conference.
- Provided technical and general user support to physicians, students, and industry reps across medical specialties.

LEADERSHIP & PROJECTS

Northwestern University Medical Makers

Evanston, IL

President

May 2025– May 2026

- Raised \$7000+ in club funding by submitting grant and departmental requests to support student experience.
- Partnered with a multinational medical device company to facilitate 6 project-specific design reviews.
- Secured a new EEG hardware company sponsorship, provided technical support to 60+ student members.

Project Manager: Waterproof Diaphragmatic Pacer

September 2024 – May 2025

- Collaborated with client, Shirley Ryan AbilityLab, to develop a waterproof diaphragmatic pacer system to allow patients to bathe with reduced nurse intervention and health risk.
- Increased attendance 3x while coordinating meetings and design challenges to boost team engagement.
- Presented progress update to 20+ nurses, earning endorsement of solution as viable in hospitals and home settings.

DSGN 384: Interdisciplinary Product Design

Evanston, IL

Primary Contributor: ENdose — Precision Liquid Medication Dosing System

September 2025 – March 2026

- Leveraged parametric OnShape CAD modeling and tolerance stack analysis to ensure manufacturability and compatibility across standard ASTM D2911 bottle finishes.
- Rendered 20+ views of the final product using SolidWorks Visualize to support client decision-making.
- Presented final design to a panel of 20+ representatives from industry and academia.